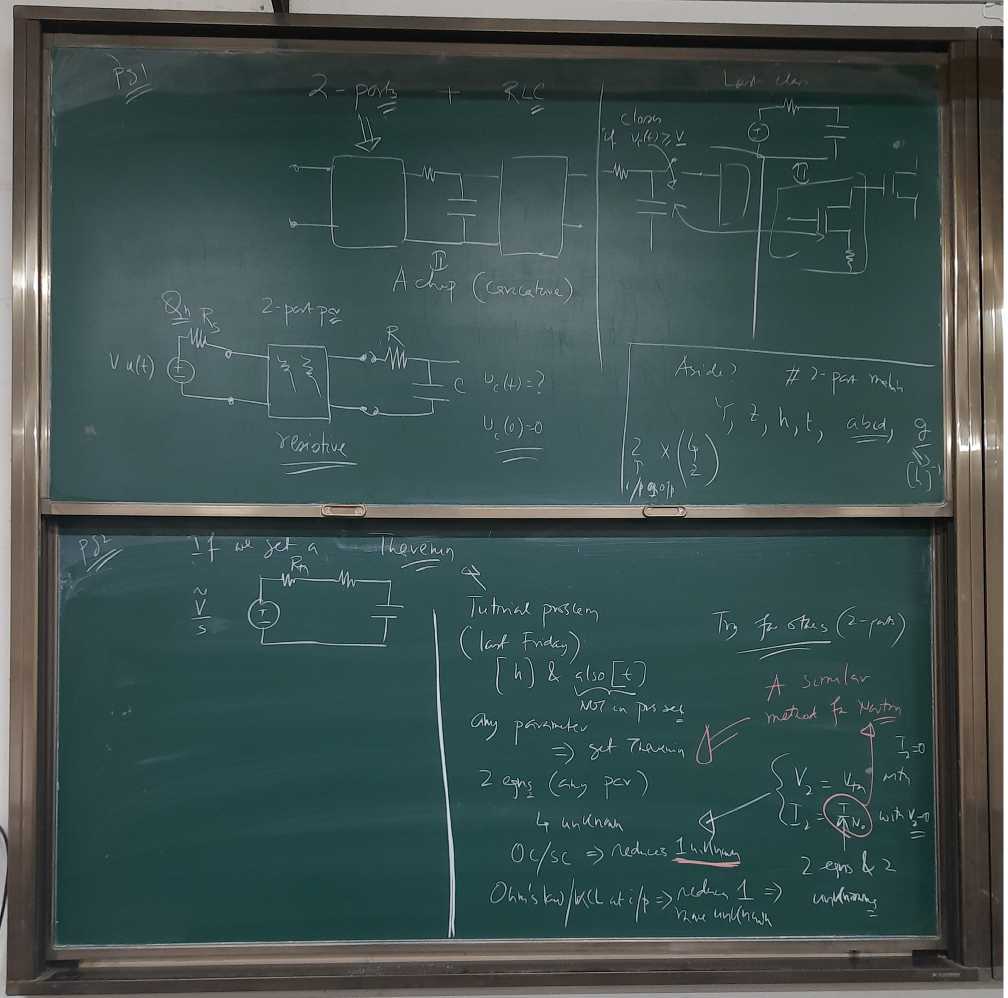

PS1
2-ports + RLC
II
A chip (caricature)
Qn
2-port par
R_s
R
$V\,u(t)$
C
resistive
Last class
clash
If $v_c(t) \geq V$
II
PS2
R_{th}
$\frac{\tilde{V}}{s}$

$v_c(t) = ?$

$v_c(0) = 0$

Aside? # 2-port matrices

Y, z, h, t, abcd, g

$(h)^{-1}$

$2 \times \binom{4}{2}$

i/p & o/p

If we get a Thevenin

Tutorial problem
(last Friday)
[h] & $\underbrace{\text{also}}_{\text{NOT in prob set}}$ [t]

any parameter
⇒ get Thevenin

2 eqns (any par)

4 unknown

OC/SC ⇒ reduces 1 unknown

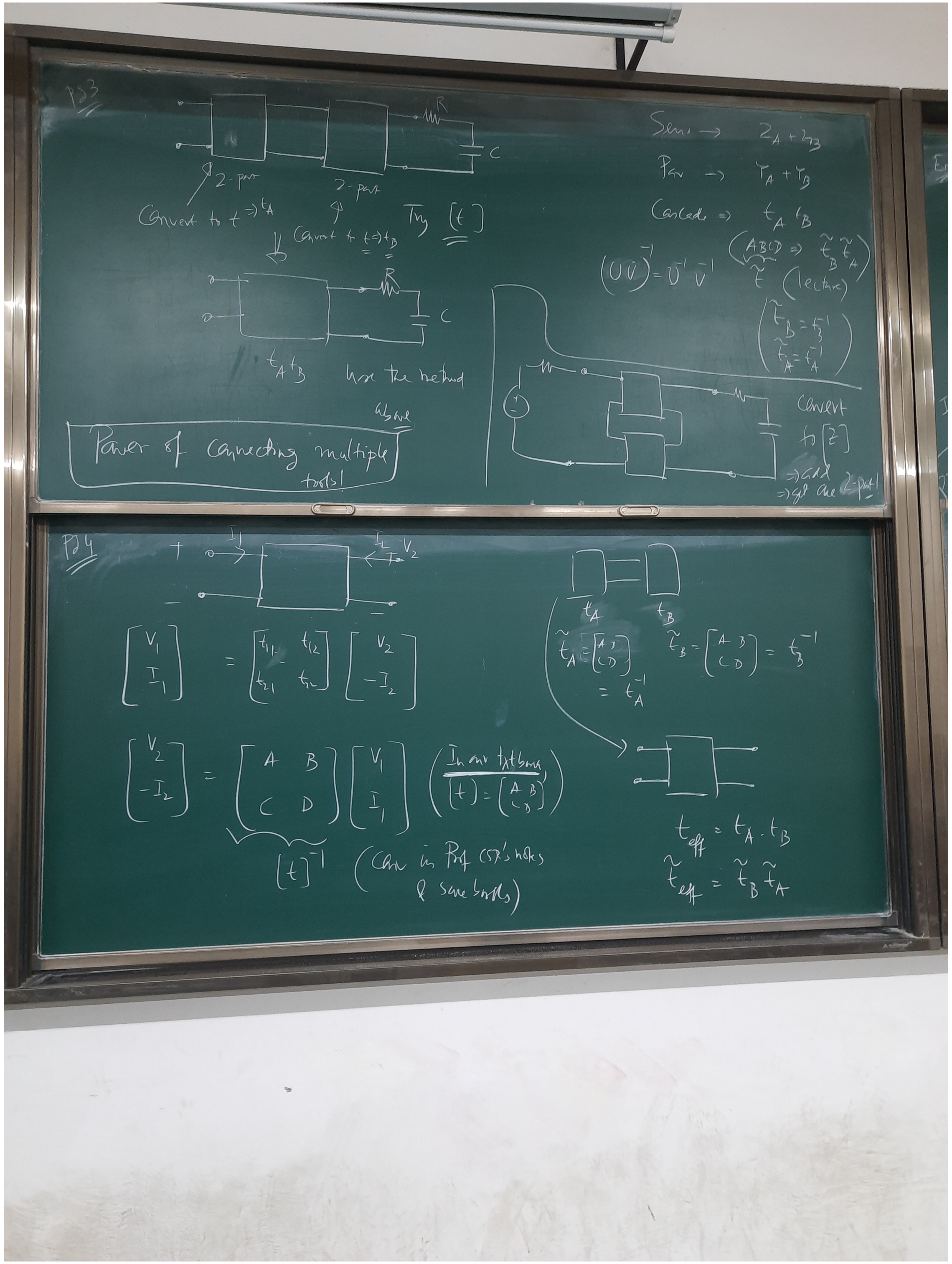
Ohm's law/KCL at i/p ⇒ reduce 1 ⇒
have unknown

Try for others (2-ports)

A similar method for Norton

$$\begin{cases} V_2 = V_{th} & \text{with } I_2 = 0 \\ I_2 = I_{No} & \text{with } V_2 = 0 \end{cases}$$

2 eqns & 2 unknowns



PS3

R
C
2-port
2-port

Convert to $t \Rightarrow t_A$

Convert to $t \Rightarrow t_B$

Try $[t]$

R
C

$t_A t_B$ Use the method above

Power of connecting multiple tools!

Series → $Z_A + Z_B$

Par → $Y_A + Y_B$

Cascade ⇒ $t_A t_B$

$\left(\underbrace{ABCD}_{\tilde{t}} \Rightarrow \tilde{t}_B \tilde{t}_A\right)$

$\tilde{t}$ (lecture)

$(UV)^{-1} = U^{-1} V^{-1}$

$\left(\begin{matrix} \tilde{t}_B = t_B^{-1} \\ \tilde{t}_A = t_A^{-1} \end{matrix}\right)$

Convert to $[Z]$
⇒ add
⇒ get one 2-port!

PS4

+
I_1
I_2
V_2
−
−

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$\begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} = \underbrace{\begin{bmatrix} A & B \\ C & D \end{bmatrix}}_{[t]^{-1}} \begin{bmatrix} V_1 \\ I_1 \end{bmatrix}$$

(In our textbook, $[t] = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$)

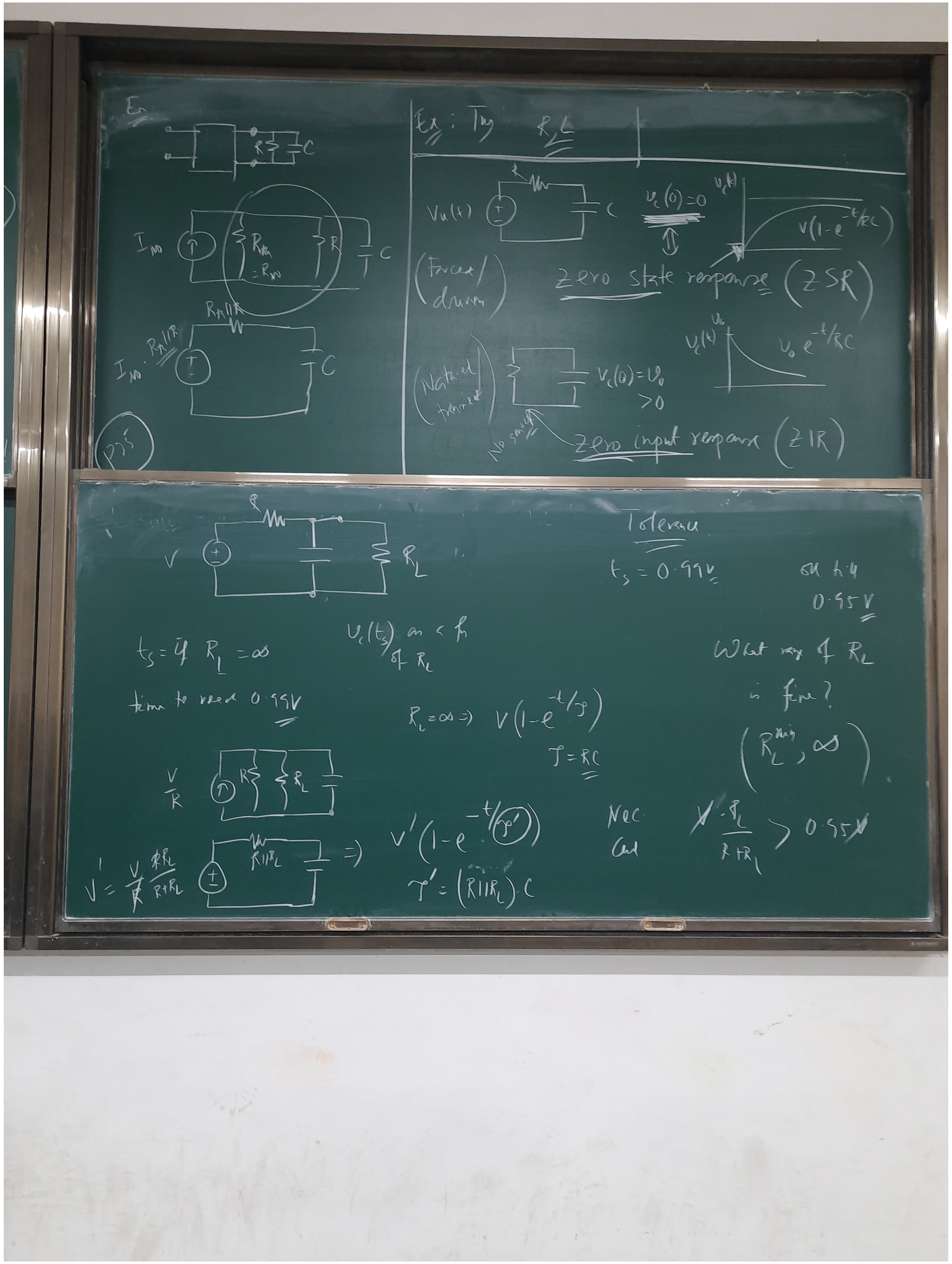
(Common in Prof CSR's notes & some books)

$\tilde{t}_A = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = t_A^{-1}$

$\tilde{t}_B = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = t_B^{-1}$

$t_{eff} = t_A \cdot t_B$

$\tilde{t}_{eff} = \tilde{t}_B \tilde{t}_A$



Ex

R C

I_{NO} $R_{Th} = R_{NO}$ R C

$R_{Th}\|R$

$I_{NO}\cdot R_{Th}\|R$ C

p75

Ex: Try $R_L C$

$V_u(t)$ R C

$v_c(0)=0$

⇕

$v_c(t)$

$V(1-e^{-t/RC})$

(Forced/driven)

Zero state response (ZSR)

$v_c(t)$ V_0

$V_0 e^{-t/RC}$

(Natural/transient)

$v_c(0)=V_0 > 0$

No source

Zero input response (ZIR)

V R R_L

$t_s = ?$ if $R_L = \infty$

time to reach 0.99V

$v_c(t_s)$ as a fn of R_L

$R_L = \infty \Rightarrow V(1-e^{-t/\tau})$

$\tau = RC$

$\frac{V}{R}$ R R_L

$V' = \frac{V}{\cancel{R}}\frac{\cancel{R}R_L}{R+R_L}$ $R\|R_L$ $\Rightarrow$ $V'(1-e^{-t/\tau'})$

$\tau' = (R\|R_L)\cdot C$

Tolerance

$t_s = 0.99V$

or till 0.95V

What range of R_L is fine?

(R_L^{min}, ∞)

Nec. Cond

$\cancel{V}\cdot\frac{R_L}{R+R_L} > 0.95\cancel{V}$